

Installation Standards – RDP housing

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Committee Chair: James Devine

Author: Jared Honey (Initial Release)

Co-Author: Johann Small (Reworked)

Approver Name	Title	Date
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James Devine	CTO	17/05/2023

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Document Index

Table of Contents

- I **ABBREVIATIONS, ACRONYMS, AND DEFINITIONS.....2**
- II **RELEVANT AND APPLICABLE DOCUMENTS.....2**
- 1. **Version History3**
- 2. **Objective3**
- 3. **Target Audience.....3**
- 4. **Introduction3**
- 5. **Installation Standards3**
 - 5.1 Entry3
 - 5.2 Cable routing6
 - 5.3 ONT/mini-UPS setup and brackets.....6
 - i. ONT/mini-UPS installation steps :7
- 6. **List of Figures12**



I ABBREVIATIONS, ACRONYMS, AND DEFINITIONS

DR	Drop
PON	Passive optical network
AC	Alternating Current
DC	Direct Current
UPS	Uninterrupted power supply
FTN	Fibertime Networks
ONT	Optical Network Terminal

II RELEVANT AND APPLICABLE DOCUMENTS

Document No	Description



1. Version History

Version	Author	Date	Comments
0.1	Jared Honey	21/07/2022	Initial release
0.2	Jared Honey	25/07/2022	Addition of mesh AP information
0.3	Godfrey Krige	04/05/2023	Addition of Mini UPS Power Solution
0.4	Johann Small	13/11/2023	Addition of formal house installation

2. Objective

This document is to be used for reference when installing the consumer equipment connecting to a network.

3. Target Audience

This document is intended for fibertime™ staff and all contractors involved in the deployment of a fibertime™ network.

4. Introduction

The installation standards have been developed by fibertime™ to all contractor's flexibility while installing consumer equipment into the consumer's home. The effort that has gone into creating this document is the result of input and collaboration from fibertime™ partners. This document should be used as a guideline within all fibertime™ networks.

- This document will cover the following aspects of the installation:
 - The entry of the fibre into a home.
 - The routing of cables, both electrical and optical.
 - The assembly of the ONT mounting bracket.
 - The mounting the ONT bracket.
 - The mounting of the Mini UPS Power Solution.

5. Installation Standards

5.1 Entry

The entry into the home should be selected using the following criteria:

- Preferably the guest bedroom, this room is closest to the pole and the Wi-Fi has signal throughout the house.
- Installation team need to advise the home-owner to install the ONT into the



guest bedroom to keep this as a standard, but if the owner wants the ONT in a different room it will be up to the supervisor to make a decision and communicate it clearly to the client.

- Should the entry point be into a non-common area, the customer must be made aware of the limitations of the Wi-Fi coverage. Factors like the thickness of walls and distance to other areas within the house must be noted to the customer should the entry point be in a non-common area.
- The entry point should be within a 10m cable run of the main power supply within the house.
- The entry point into the house must be as close as possible to the ONT location on the inside of the house.
- Preferably land on the roof trusses behind the fascia board by drilling a hole through the fascia board into the roof trusses.



Figure 1: Example of entry into an RDP home

- Use a pigtail screw and screw into the pilot hole.
- Secure the drop cable by wrapping a dead-end onto the drop cable and hooking it to the pigtail screw.





Figure 2: Secure drop cable to pigtail screw

- Drill a 12mm hole through the wall of the house.
- Insert a 12/10mm duct, the same length as the wall that was drilled through, to secure the drop.
- Make sure the duct has been cut from end to end to slide over the drop cable because the SC-APC connector does not fit through the duct.



Figure 3: Hole in the wall for feeding the drop cable through

- Seal the duct after drop cable has been installed.
- If there was any damages to the wall due to drilling, this needs to be fixed by using Polyfilla.



5.2 **Cable routing**

Fibertime™ has a high standard for neatness and all cable runs will follow the following criteria.

- Cable runs are to be as unintrusive as possible and thusly cable runs shall be limited to natural contours of the structure such as corners of walls, along skirting or ceilings.
- Cables runs in open space on walls shall only be utilised as a last resort.
- Cable runs within the house should be as short as possible.
- All cable slack from the installation should be neatly hidden behind the fibertime™ Optical Network Terminal (ONT) or the mounting bracket.
- The power supply cable of the Gizzu (UPS) must be secured to the wooden board and neatly routed to the main power supply of the house by using a glue gun or saddles along the skirtings.



Figure 4: A neat cable run in an informal home

5.3 **ONT/mini-UPS setup and brackets**

The ONT/mini-UPS setup consists of the following;

1. Varnished Wooden Board (280mm x 120mm) with 4 pre-drilled holes.
2. ONT bracket.
3. X 2 flat head, self-tapping screws for the ONT bracket.
4. X 2 self-tapping screws or fisher plugs for bricks for the Varnished Wooden Board (280mm x 120mm).
5. X 1 black cable tie for mounting the Gizzu.
6. Nokia Optical Network Terminal (ONT).
7. Mini-UPS (Gizzu).
8. 10m 3 prong Power supply cord for the Gizzu.



9. Multi plug.
10. Fibertime™ ONT sticker.
11. Brady Tape to enter Drop Number that matches drop label at distribution point.

i. ONT/mini-UPS installation steps :

1. Varnished Wooden board (280mm x 120mm)

- The wooden board needs to be varnished and the edges need to be trimmed.
- Fix the wooden board against the wall by using a level. When level, mark the 2 holes and drill 2 holes into the wall with a 6mm bit.
- Tighten the wooden board to the wall using x 2 self-tapping screws.



Figure 5: A varnished wooden board for ONT installation

2. ONT bracket

- Drill a 5mm hole in the middle of the ONT bracket for the cable tie to go through.
- Place the ONT bracket in the middle of the wooden board, mark the 2 holes and fix it to the board with x 2 flat head, self-tapping screws. Make sure a level has been used before the screws are tightened.





Figure 6: An ONT bracket installed in an RDP home

3. Drop cable slack

- Coil the drop cable slack (no more than 10m) to a loop of no bigger than 300mm diameter. This slack is to prevent breaking the fiber when the ONT is unmounted for maintenance.
- Use 2x black cable ties on the loop to keep it neat.
- Put the drop cable slack over the ONT bracket before installing the mini-UPS.



Figure 7: An indication of the placement of the drop cable slack

4. Mini-UPS (Gizzu) installation

- Remove the plastic tape that is on the middle of the Gizzu.
- Mount the mini-UPS power solution on top of the ONT bracket. Secure the mini-UPS by tightening it to the ONT bracket with a cable tie.
- Plug the supplied multi plug into the legal power supply of the house.



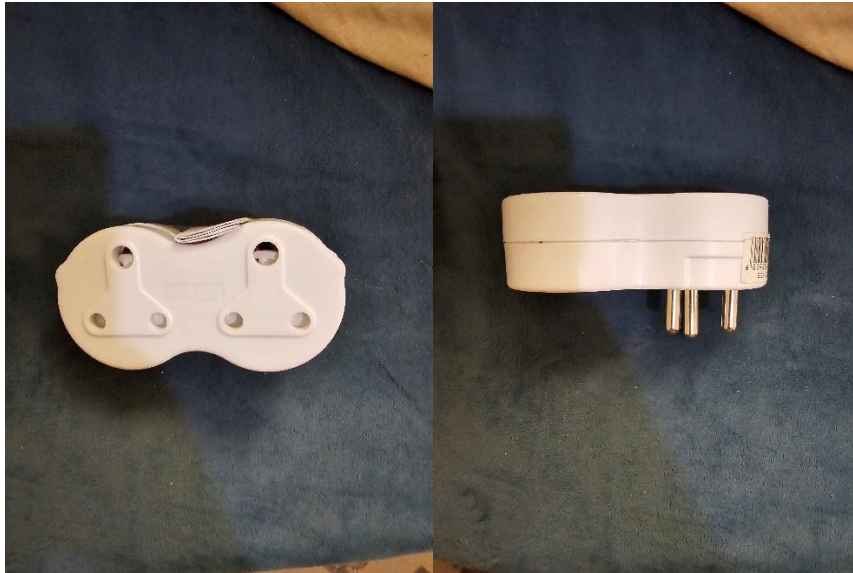


Figure 8: Supplied multiplug for use with GIZZU

- Plug the AC power cable of the mini-UPS Power solution into the AC input socket of the house and the other end in the type D plug of the mini-UPS.
- Once this is done, and the extension cable is plugged in at the wall and switched on, the battery lights of the mini-UPS power solution will light up (this is not the case during loadshedding).
- Connect the DC output cable (2 ends) to the output socket of the mini-UPS power solution and the other end to the ONT DC jack port when installing the ONT.
- The output voltage selector on the mini-UPS power solution must be on 12V.
- Secure the power cable of the mini-UPS to the wooden board by using a glue gun or saddles.



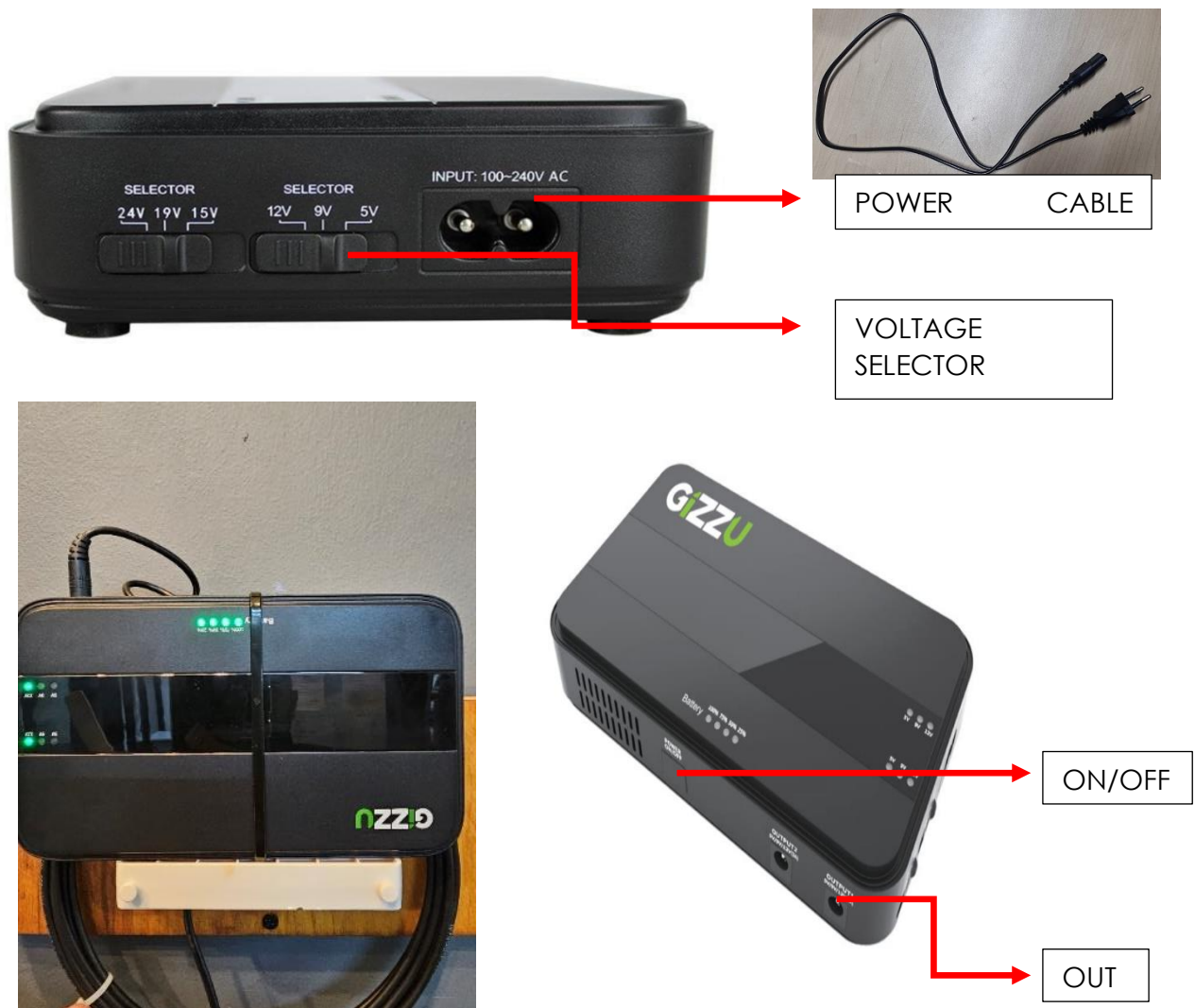


Figure 9: Schematics regarding the Mini-UPS (GIZZU) installation

5. ONT installation

- Fiber needs to be coiled anti-clockwise at the back of the ONT to be able to enter the PON port without making any bends.
- Connect the DC output cable of the mini-UPS into the DC jack port of the ONT.
- Install the ONT onto the ONT bracket by sliding the ONT mounting holes into the ONT bracket.
- The antennas of the ONT must be in the upright position.
- Neaten the DC cable by using a small cable tie and store it behind the ONT.



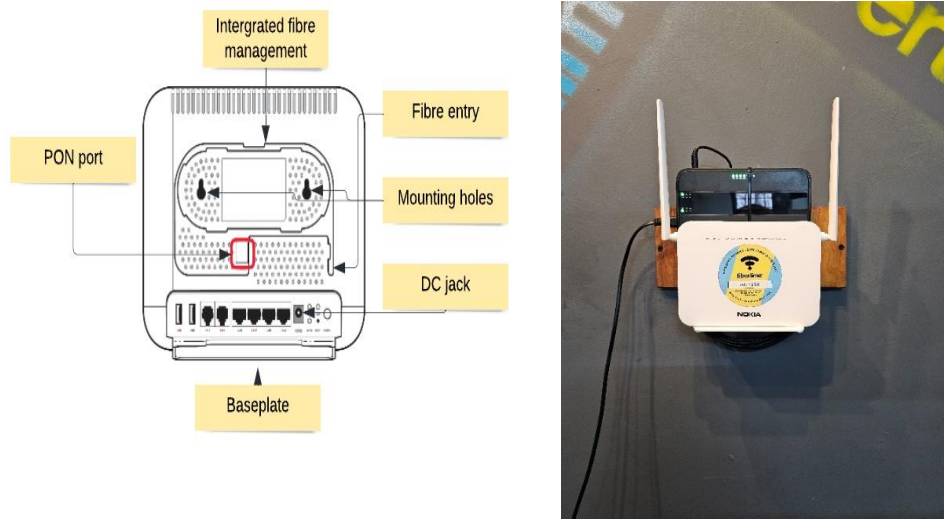


Figure 10: Schematics regarding the ONT installation

6. Fibertime™ sticker

- Apply the sticker onto the middle of the ONT. Make sure it is as straight and neat as possible and none of the lights are covered.
- Print the drop number according to Fibertime™ naming standards.
➤ DRXXXX
- Apply the drop number label onto the sticker where the space is provided.
- The drop label must be black on yellow.
- Apply the small Serial number sticker of the ONT below the Nokia name.

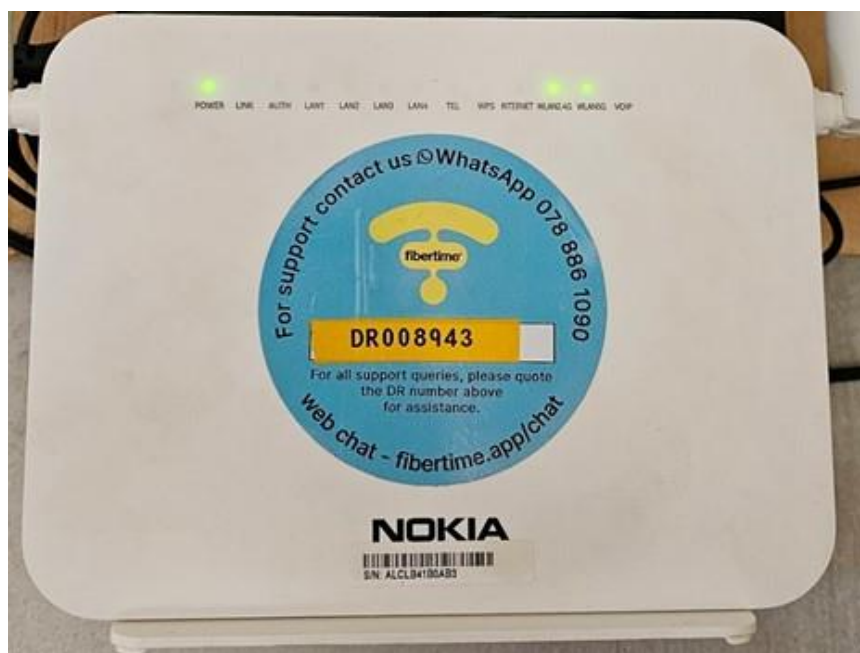


Figure 11: Placement of the fibertime and Nokia stickers



7. Installed ONT/mini-UPS

- The final product needs to look like the attached photos.

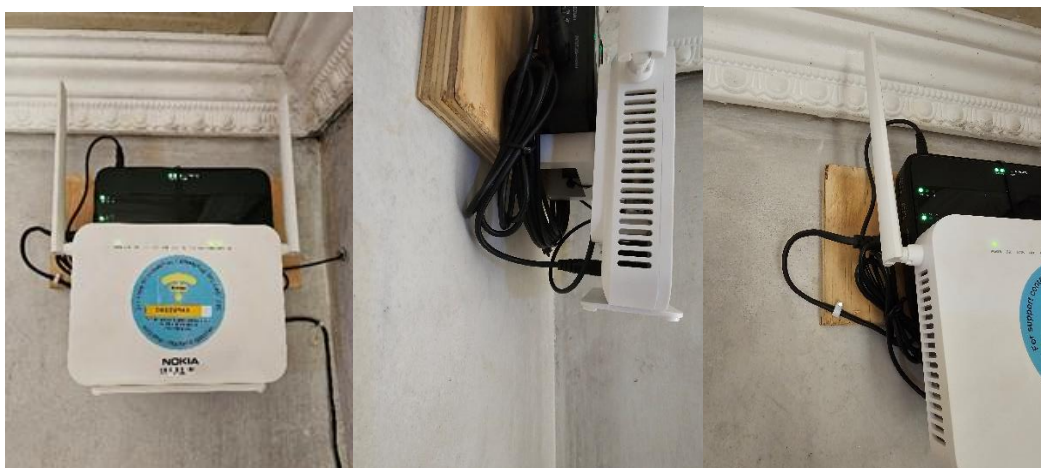


Figure 12: Final installation of ONT and mini-UPS

6. List of Figures

Figure 1: Example of entry into an RDP home	4
Figure 2: Secure drop cable to pigtail screw	5
Figure 3: Hole in the wall for feeding the drop cable through	5
Figure 4: A neat cable run in an informal home	6
Figure 5: A varnished wooden board for ONT installation	7
Figure 6: An ONT bracket installed in an RDP home	8
Figure 7: An indication of the placement of the drop cable slack	8
Figure 8: Supplied multiplug for use with GIZZU	9
Figure 9: Schematics regarding the Mini-UPS (GIZZU) installation.....	10
Figure 10: Schematics regarding the ONT installation.....	11
Figure 11: Placement of the fibertime and Nokia stickers	11
Figure 12: Final installation of ONT and mini-UPS.....	12

